

Security Log Investigation: Project Solution Guide

Objective: Turn raw event records into defensible detection findings and response actions.

1. Project Objectives & Deliverables

This document provides a comprehensive structured template and guide to complete the Security Log Investigation task successfully. To pass the requirements, your submission must address the following areas:

- **Investigation Notebook or Spreadsheet:** Documenting the technical steps taken.
- **Incident Report PDF:** Summarizing findings for leadership or stakeholders.

2. Step-by-Step Implementation Guide

Step 2.1: Data Normalization

Raw logs are often disorganized. You must clean and parse the data into a standard tabular format consisting of the following key fields:

- **Timestamp:** Convert all log dates and times to a consistent timezone (e.g., UTC) in standard format (YYYY-MM-DD HH:MM:SS).
- **IP Addresses:** Parse Source and Destination IPs, filtering out invalid strings or noise.
- **User Accounts:** Map inconsistent identity values into single standardized usernames.
- **Actions & Results:** Categorize operations (e.g., LOGIN, LOGOUT, FILE_ACCESS) and outcomes (e.g., SUCCESS, FAILURE).

Step 2.2: Defining Detection Rules

Implement logic queries (using SQL, Python/Pandas, or Spreadsheet Filters) to discover indicators of compromise (IoCs):

Attack Type / Scenario	Detection Rule & Logic Criteria
Credential Spraying	Identify a single source IP attempting authentication across multiple distinct user accounts within a very short duration (e.g., < 5 minutes) resulting in mostly failures.
Brute Force Attack	Identify a high volume of failed login attempts targeting a single specific user account from one or more source IPs, typically followed by a single successful login entry.
Unusual / Anomalous Access	Detect successful authentications originating from unexpected geographical locations, unfamiliar hosting providers, or occurring during anomalous hours relative to historical user baselines.

Step 2.3: Timeline Construction & Evidence Separation

Organize anomalies in strict chronological sequence to trace the attacker's trajectory:

- **Chronological Order:** Arrange events systematically from initial reconnaissance/spraying, to successful breach, to subsequent privilege escalation or data exfiltration.
- **Facts vs. Assumptions:** Always distinguish concrete proof from working hypotheses. A recorded failure log from an IP is a *Fact*; concluding that the specific IP belongs to a targeted nation-state without external threat intelligence is an *Assumption*.

3. Final Submission Requirements

Ensure that your public repository or shared folder contains all operational notebooks, raw datasets, and reports. Verify that your sharing settings are configured as '**Anyone with the link can view**' before submitting the single required proof link to pass.